

phi-2

Beta

Text Generation • Microsoft

@cf/microsoft/phi-2

Phi-2 is a Transformer-based model with a next-word prediction objective, trained on 1.4T tokens from multiple passes on a mixture of Synthetic and Web datasets for NLP and coding.

Model Info

Context Window ↗	2,048 tokens
More information	link ↗
Beta	Yes

Playground

Try out this model with Workers AI LLM Playground. It does not require any setup or authentication and an instant way to preview and test a model directly in the browser.

Launch the LLM Playground

Usage

▼ Worker - Streaming

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const stream = await  
env.AI.run("@cf/microsoft/phi-2", {  
      messages,  
      stream: true,  
    });
```

```
return new Response(stream, {  
  headers: { "content-type":  
"text/event-stream" },  
  });  
},  
} satisfies ExportedHandler<Env>;
```

▼ Worker

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const response = await  
env.AI.run("@cf/microsoft/phi-2", { messages  
});  
  
    return Response.json(response);  
  },  
} satisfies ExportedHandler<Env>;
```

▼ Python

Python

```
import os
import requests

ACCOUNT_ID = "your-account-id"
AUTH_TOKEN = os.environ.get("CLOUDFLARE_AUTH_TOKEN")

prompt = "Tell me all about PEP-8"
response = requests.post(
    f"https://api.cloudflare.com/client/v4/accounts/{ACCOUNT_ID}/ai/v1/chat/completions",
    headers={"Authorization": f"Bearer {AUTH_TOKEN}"},
    json={
        "messages": [
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt}
        ]
    }
)
result = response.json()
print(result)
```

▼ curl

```
curl
https://api.cloudflare.com/client/v4/accounts/
2 \
-X POST \
-H "Authorization: Bearer $CLOUDFLARE_AUTH_TOKEN" \
-d '{ "messages": [{ "role": "system", "content": "You are a helpful assistant." }, { "role": "user", "content": "Why is pizza so good" } ] }'
```

① OpenAI compatible endpoints

Workers AI also supports OpenAI compatible API endpoints for `/v1/chat/completions` and `/v1/embeddings`. For more details, refer to [Configurations](#).

Parameters

* indicates a required field

Input

• **object**

- **prompt** **string** required min 1

The input text prompt for the model to generate a response.

- **lora** **string**

Name of the LoRA (Low-Rank Adaptation) model to fine-tune the base model.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0.001 max 1

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- **seed** **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- **repetition_penalty** **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- **frequency_penalty** **number** min -2 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- **presence_penalty** **number** min -2 max 2

Increases the likelihood of the model introducing new topics.

- ▶ **1** **object**

- **messages** **array** required

An array of message objects representing the conversation history.

- **items** **object**

- **role** **string** required

The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- **content** **string** required

The content of the message as a string.

- **functions** **array**

- **items** **object**

- **name** **string** required

- **code** **string** required

- **tools** **array**

A list of tools available for the assistant to use.

- **items** **one of**

- **0** **object**

- **name** **string** required

The name of the tool. More descriptive the better.

- **description** **string** required

A brief description of what the tool does.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

■ 1 **object**

- **type** **string** required

Specifies the type of tool (e.g., 'function').

- **function** **object** required

Details of the function tool.

- **name** **string** required

The name of the function.

- **description** **string** required

A brief description of what the function does.

- **parameters** **object** required

Schema defining the parameters accepted by the function.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **max_tokens** **integer** default 256

The maximum number of tokens to generate in the response.

- **temperature** **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- **top_p** **number** min 0.001 max 1

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min -2 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min -2 max 2

Increases the likelihood of the model introducing new topics.

Output

• **0** **object**

- response **string** required

The generated text response from the model

- usage **object**

Usage statistics for the inference request

- prompt_tokens **number** 0

Total number of tokens in input

- completion_tokens **number** 0

Total number of tokens in output

- total_tokens **number** 0

Total number of input and output tokens

- tool_calls **array**

An array of tool calls requests made during the response generation

- items **object**

- arguments **object**

The arguments passed to be passed to the tool call request

- name **string**

The name of the tool to be called

- 1 **string**

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{  
  "type": "object",  
  "oneOf": [  
    {  
      "title": "Prompt",  
      "properties": {  
        "prompt": {  
          "type": "string",  
          "minLength": 1,  
          "description": "The input  
text prompt for the model to generate a  
response."  
        },  
        "lora": {  
          "type": "string",  
          "description": "Name of  
the LoRA (Low-Rank Adaptation) model to fine-  
tune the base model."  
        },  
        "response_format": {  
          "title": "JSON Mode",  
          "type": "object",  
          "properties": {  
            "type": {  

```

```
        "type": "string",
        "enum": [
            "json_object",
            "json_schema"
        ]
    },
    "json_schema": {}
},
"raw": {
    "type": "boolean",
    "default": false,
    "description": "If true, a
chat template is not applied and you must
adhere to the specific model's expected
formatting."
},
"stream": {
    "type": "boolean",
    "default": false,
    "description": "If true,
the response will be streamed back
incrementally using SSE, Server Sent Events."
},
"max_tokens": {
```



```
    "type": "integer",  
    "default": 256,  
    "description": "The  
maximum number of tokens to generate in the  
response."  
  },  
  "temperature": {  
    "type": "number",  
    "default": 0.6,  
    "minimum": 0,  
    "maximum": 5,  
    "description": "Controls  
the randomness of the output; higher values  
produce more random results."  
  },  
  "top_p": {  
    "type": "number",  
    "minimum": 0.001,  
    "maximum": 1,  
    "description": "Adjusts  
the creativity of the AI's responses by  
controlling how many possible words it  
considers. Lower values make outputs more  
predictable; higher values allow for more  
varied and creative responses."
```

```
    },  
    "top_k": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 50,  
      "description": "Limits the  
AI to choose from the top 'k' most probable  
words. Lower values make responses more  
focused; higher values introduce more variety  
and potential surprises."  
    },  
    "seed": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 9999999999,  
      "description": "Random  
seed for reproducibility of the generation."  
    },  
    "repetition_penalty": {  
      "type": "number",  
      "minimum": 0,  
      "maximum": 2,  
      "description": "Penalty  
for repeated tokens; higher values discourage  
repetition."
```

```
    },  
    "frequency_penalty": {  
      "type": "number",  
      "minimum": -2,  
      "maximum": 2,  
      "description": "Decreases  
the likelihood of the model repeating the same  
lines verbatim."  
    },  
    "presence_penalty": {  
      "type": "number",  
      "minimum": -2,  
      "maximum": 2,  
      "description": "Increases  
the likelihood of the model introducing new  
topics."  
    }  
  },  
  "required": [  
    "prompt"  
  ]  
},  
{  
  "title": "Messages",  
  "properties": {
```

```
"messages": {  
  "type": "array",  
  "description": "An array  
of message objects representing the  
conversation history.",  
  "items": {  
    "type": "object",  
    "properties": {  
      "role": {  
        "type":  
"string",  
        "description":  
"The role of the message sender (e.g., 'user',  
'assistant', 'system', 'tool')."  
      },  
      "content": {  
        "type":  
"string",  
        "description":  
"The content of the message as a string."  
      }  
    },  
    "required": [  
      "role",  
      "content"
```

```
    ]
  }
},
"functions": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "name": {
        "type":
"string"

      },
      "code": {
        "type":
"string"

      }
    },
    "required": [
      "name",
      "code"
    ]
  }
},
"tools": {
  "type": "array",
```

```
    "description": "A list of
tools available for the assistant to use.",
    "items": {
      "type": "object",
      "oneOf": [
        {
          "properties":
{
            "name": {
              "type": "string",

"description": "The name of the tool. More
descriptive the better."
            },

"description": {
              "type": "string",

"description": "A brief description of what
the tool does."
            },

"parameters": {
```

```
"type": "object",

"description": "Schema defining the parameters
accepted by the tool.",

"properties": {

  "type": {

    "type": "string",

    "description": "The type of the parameters
object (usually 'object')."

    },

    "required": {

      "type": "array",

      "description": "List of required parameter
names.",

      "items": {
```

```
"type": "string"
```

```
}
```

```
},
```

```
"properties": {
```

```
"type": "object",
```

```
"description": "Definitions of each  
parameter.",
```

```
"additionalProperties": {
```

```
"type": "object",
```

```
"properties": {
```

```
"type": {
```

```
"type": "string",
```

```
"description": "The data type of the  
parameter."
```



```
},  
  
"description": {  
  
"type": "string",  
  
"description": "A description of the expected  
parameter."  
  
}  
  
},  
  
"required": [  
  
"type",  
  
"description"  
  
]  
  
}  
  
},
```

```
"required": [  
  
  "type",  
  
  "properties"  
  
],  
},  
"required": [  
  "name",  
  
  "description",  
  
  "parameters"  
  
],  
{  
  "properties":  
  
    {  
  
      "type": {  
  
        "type": "string",  
  
        "description": "Specifies the type of tool"
```

```
(e.g., 'function')."
```

```
},
```

```
"function": {
```

```
"type": "object",
```

```
"description": "Details of the function  
tool.",
```

```
"properties": {
```

```
"name": {
```

```
"type": "string",
```

```
"description": "The name of the function."
```

```
},
```

```
"description": {
```

```
"type": "string",
```

```
"description": "A brief description of what  
the function does."
```

},

```
"parameters": {  
  
  "type": "object",  
  
  "description": "Schema defining the parameters  
accepted by the function.",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "The type of the parameters  
object (usually 'object')."  
  
    },  
  
    "required": {  
  
      "type": "array",  
  
      "description": "List of required parameter
```

```
names.",  
  
"items": {  
  
"type": "string"  
  
}  
  
},  
  
"properties": {  
  
"type": "object",  
  
"description": "Definitions of each  
parameter.",  
  
"additionalProperties": {  
  
"type": "object",  
  
"properties": {  
  
"type": {
```

```
"type": "string",

"description": "The data type of the
parameter."

},

"description": {

"type": "string",

"description": "A description of the expected
parameter."

}

},

"required": [

"type",

"description"
```

```
]
```

```
}
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"properties"
```

```
]
```

```
}
```

```
},
```

```
"required": [
```

```
"name",
```

```
"description",
```

"parameters"

```
    ]
  },
  "required": [
    "type",
    "function"
  ]
}
]
}
},
"response_format": {
  "title": "JSON Mode",
  "type": "object",
  "properties": {
    "type": {
      "type": "string",
      "enum": [
        "json_object",
        "json_schema"
      ]
    },
  },
  "json_schema": {}
}
```



```
    }  
  },  
  "raw": {  
    "type": "boolean",  
    "default": false,  
    "description": "If true, a  
chat template is not applied and you must  
adhere to the specific model's expected  
formatting."  
  },  
  "stream": {  
    "type": "boolean",  
    "default": false,  
    "description": "If true,  
the response will be streamed back  
incrementally using SSE, Server Sent Events."  
  },  
  "max_tokens": {  
    "type": "integer",  
    "default": 256,  
    "description": "The  
maximum number of tokens to generate in the  
response."  
  },  
  "temperature": {
```

```
    "type": "number",
    "default": 0.6,
    "minimum": 0,
    "maximum": 5,
    "description": "Controls
the randomness of the output; higher values
produce more random results."
  },
  "top_p": {
    "type": "number",
    "minimum": 0.001,
    "maximum": 1,
    "description": "Adjusts
the creativity of the AI's responses by
controlling how many possible words it
considers. Lower values make outputs more
predictable; higher values allow for more
varied and creative responses."
  },
  "top_k": {
    "type": "integer",
    "minimum": 1,
    "maximum": 50,
    "description": "Limits the
AI to choose from the top 'k' most probable
```

words. Lower values make responses more focused; higher values introduce more variety and potential surprises."

```
    },  
    "seed": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 9999999999,  
      "description": "Random  
seed for reproducibility of the generation."  
    },  
    "repetition_penalty": {  
      "type": "number",  
      "minimum": 0,  
      "maximum": 2,  
      "description": "Penalty  
for repeated tokens; higher values discourage  
repetition."  
    },  
    "frequency_penalty": {  
      "type": "number",  
      "minimum": -2,  
      "maximum": 2,  
      "description": "Decreases  
the likelihood of the model repeating the same
```

```
lines verbatim."
    },
    "presence_penalty": {
      "type": "number",
      "minimum": -2,
      "maximum": 2,
      "description": "Increases
the likelihood of the model introducing new
topics."
    }
  },
  "required": [
    "messages"
  ]
}
]
```

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